End-to-End MERN Application Architecture & Best Practices

Name: Zedekiah Heteroza

Student ID: 24-0521-842

CHALLENGES

Challenges Encountered and Solutions

During the development of the MERN lab project, I encountered several challenges related to project setup, MongoDB connection, React configuration, and API testing.

One challenge I encountered was a MongoDB connection error involving unsupported options such as useNewUrlParser and useUnifiedTopology. This issue happened because the installed version of Mongoose no longer required those options. I resolved the issue by removing the unsupported options from the database connection code in db.js.

Another issue occurred when running the React frontend. The application displayed the error “Could not find a required file: index.js”. I discovered that the src/index.js file was missing. I fixed the problem by recreating the required index.js file and ensuring the React project structure remained complete.

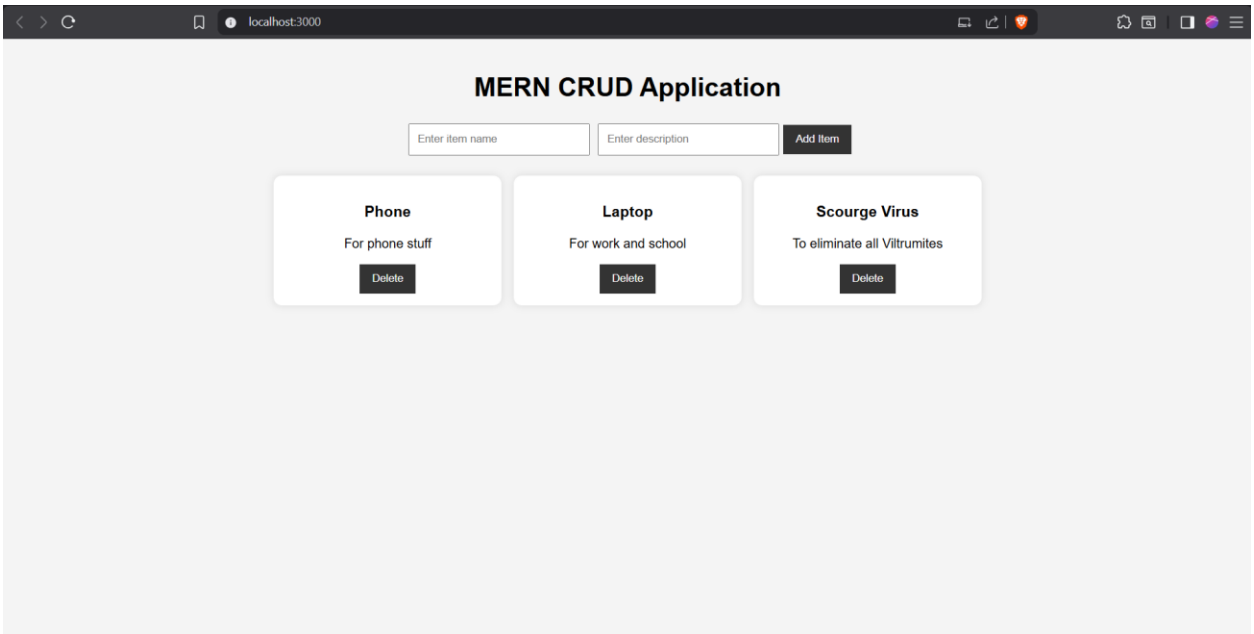
I also encountered validation errors while testing API routes in Postman. One example was the “Cast to ObjectId failed” error when creating a task document. This happened because I used a placeholder string instead of a valid MongoDB ObjectId. I resolved this by first creating a user document and then copying the generated _id into the userId field when creating tasks.

Another challenge was organizing the project structure and understanding how backend and frontend components communicate in a MERN application. I improved the structure by separating the project into folders such as models, routes, config, and components to make the code cleaner and easier to maintain.

Through this activity, I learned how to:

- Connect MongoDB using Mongoose
- Build REST API routes using Express
- Perform CRUD operations
- Use React components and state
- Debug common MERN stack errors

- Improve project organization and readability

Overall, the lab helped me better understand the MERN stack workflow and debugging process.

SCREENSHOT OF FRONTEND